



SawtArabi: A Benchmark Corpus for Arabic TTS. Standard, Dialectal and Code-Switching

Vasista Sai Lodagala^{1,*}, Lamya Alkanhal^{2,*}, Daniel Izham¹, Shivam Mehta³, Shammur Chowdhury⁴,
Aqeelah Makki¹, Hamdy S. Hussein⁴, Gustav Eje Henter³, Ahmed Ali¹

¹Humain, Saudi Arabia

²Saudi Data AI Authority, Saudi Arabia

³KTH Royal Institute of Technology, Sweden

⁴QCRI, Qatar

vasista.lodagala@gmail.com, lalkanhal@nic.gov.sa, mizham@thefutureai.co, smehta@kth.se,
shchowdhury@hbku.edu.qa, amakki@thefutureai.co, hmubarak@hbku.edu.qa, ghe@kth.se,
ahmed.email.ali@ieee.org

Abstract

Curating Text-to-Speech (TTS) datasets is a strenuous task given the quality considerations. While it is hard to find high-quality TTS datasets in languages other than English, it is rare to come across code-switching (CS) datasets. As a part of this work, we curate a 4-hour Arabic-English TTS corpus consisting of code-switched Egyptian-English, monolingual Modern Standard Arabic (MSA), Egyptian, and English, all recorded by the same voice talent.

We demonstrate the importance of vowelization and the need for better phonemization of Arabic text. To this effect, we present the modified *espeak-ng* phonemizer that handles various irregularities of *espeak-ng* over Arabic text. Upon training baseline TTS systems over this benchmark, we demonstrate its efficacy through extensive subjective evaluations.

Index Terms: Text-to-Speech Synthesis, Code-switching, Multilingual, Dialectal Speech, Phonemization

1. Introduction and Related Work

Great progress in machine learning, deep learning and speech large language modeling approaches have lowered the barrier in build high quality Text-to-Speech (TTS) systems. These achievements are largely attributed to the scale-up of training data and the strong modeling capabilities of the models. For example, VALL-E [1] scaled up its training data to 60,000 hours of English speech [2] and NaturalSpeech [3] leveraged over 44,000 hours of transcribed English speech data [4]. Recently, WenetSpeech4TTS [5] released over 12,800 hours in a multi-domain Mandarin corpus derived from the open-source WenetSpeech dataset. Also, given the complexity in gathering code-switching datasets, efforts have been made in addressing code-switching TTS at a modeling level by either using bilingual data [6, 7] or by using cross-lingual word embeddings [8] or voice conversion abilities [9]. To this end, there is a noticeable lack of extensive open-source datasets for Arabic TTS applications compared to English or multilingual datasets [10].

Moreover, limited research and data are available for Arabic TTS. Here, we review previous work: (i) CIArTTS [11], a 12-hour single-speaker Classical Arabic corpus. Extracted from a LibriVox audiobook, it was transcribed and vowelized manually. Proposed models show CIArTTS improves TTS quality over existing corpora, and it is publicly available for research; (ii) ARTST [12], a pre-trained Arabic text and speech transformer, fine-tuned for ASR, TTS, and Dialect Identification. ARTST

outperforms multilingual models like Whisper and MMS and learns TTS without explicit diacritization. The model and fine-tuned systems are publicly available for research. Such pre-trained models can perform multiple downstream speech-related tasks with different input modalities; (iii) NatiQ [13], an end-to-end Arabic TTS system leveraging Transformer and Parallel WaveGAN [14], achieving MOS scores of 4.40 (neutral voice) and 4.21 (expressive). With a real-time factor of 0.09 on GPU, the models have been trained over male and female voice talents for MSA only. Data and models are not public.

Earlier works have addressed MSA speech, using high resource and clean recordings. (iv) Finally, a recent study by [10] proposed an unsupervised method for low-resource Arabic TTS using broadcast news, combining automatic data selection (DNSMOS), human/automatic vowelization, and transfer learning. This approach enables efficient TTS development for under-resourced languages, validated via Arabic Aljazeera data.

Previous work lacks a standardized benchmark for Arabic TTS and is not tailored to the unique characteristics of the Arabic language. Key gaps include the absence of dialectal TTS, which is essential for capturing Arabic's linguistic diversity, and the lack of code-switching TTS, crucial for handling mixed-language speech common in everyday conversations.

The major contributions from this work are the following:

- Single-speaker voice recordings in Egyptian Arabic, MSA, English, and Egyptian-English code-switching. To the best of our knowledge, this is the first work to address dialectal and code-switching TTS in the Arabic language context.
- We propose modifications to the existing *espeak-ng* phonemizer that is widely used in training TTS systems. Certain rules for Arabic have been modified to result in an accurate phonemization of the text. This in turn aids TTS system training, as validated by our study.
- We demonstrate the significance of vowelization of Arabic text while building TTS systems.
- Considering the dialects and code-switching involved, we believe that this dataset would act as a benchmark for further TTS training and evaluation in Arabic. Given that the speaker is a native Egyptian speaker, the data from this study also helps to analyze the characteristics of L2-speaking over other dialects and code-switching data.

The SawtArabi benchmark corpus, the modified *espeak-ng* phonemizer and the baseline checkpoints along with their associated code-base are available for public access ¹.

*These authors contributed equally.

¹<https://huggingface.co/spaces/ArabicSpeech/sawtarabi>

Table 1: *Dataset Statistics: The average number of characters, average number of words and the average duration are calculated per utterance in the dataset. En-Ar stands for English-to-Arabic*

Metric	Code-Switching (CS)			Egyptian (EGY)			MSA			English (ENG)		
	train	dev	test	train	dev	test	train	dev	test	train	dev	test
Number of Utterances	687	132	65	672	143	76	486	111	58	659	144	71
Duration (mm:ss)	53:44	09:58	04:59	45:00	09:58	05:01	42:36	10:06	04:58	44:58	09:58	05:02
Avg. number of characters	69.4	66.1	65.9	60.2	63.0	58.7	68.5	71.5	67.9	42.6	42.3	43.4
Avg. number of words	9.8	9.4	9.5	8.9	9.2	8.7	8.5	8.5	8.0	8.4	8.4	8.4
Avg. duration (seconds)	4.7	4.5	4.6	4.0	4.2	4.0	5.3	5.4	5.1	4.0	4.2	4.3
En-Ar character ratio	0.47	0.51	0.57	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A
En-Ar word ratio	0.38	0.42	0.45	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A	N.A

2. SawtArabi Benchmark

The word ‘‘Sawt’’ in Arabic stands for ‘‘Voice’’. Our efforts as a part of this work are to create a benchmarking TTS dataset for MSA, dialectal and code-switching Arabic. The next subsections detail the curation and characteristics of this benchmark.

2.1. Text Sources and transcription guidelines

SawtArabi text has been collected from varied sources depending on the dialect or language. MSA text is a subset of the NatiQ data after contacting its authors [13]; we ensured that the text was fully MSA, avoiding any dialectal or classical Arabic. English data has been randomly selected from LJspeech [15]. However, the speaker (being a native Egyptian) struggled multiple times to read certain English sentences; therefore, he was allowed to edit or replace difficult words (e.g., replacing *difficult* with *hard*). Finally, Egyptian and code-switching texts have been generated using GPT-4 LLM [16]. For code-switching, we ensured that the Arabic text primarily followed the Egyptian dialect.

The text given to the professional speaker had no vowelization, and the speaker had the freedom to make any changes that made recording easier, as long as they were minor edits. Therefore, the text had to be reviewed to ensure the transcription was fully correct and the pronounced vowelization has been applied, even if it was linguistically incorrect. For Egyptian and code-switching data, we had to address the fact that dialectal speech, in many cases, does not follow the standard orthography. The speaker might pronounce تلاط (t $\text{\AA}$ Avp²) as تلاط (t $\text{\AA}$ Atp). Our approach was to write the text as it has been pronounced, not as it should have been written, which is a reasonable assumption as long as the TTS system is dealing with one dialect at a time and not mixing dialects, for dialectal CS, such as EGY with MSA [17], which is a common occurrence in formal speech.

2.2. Dataset Statistics

The entire SawtArabi dataset has been recorded by one professional speaker (male aged 34 years) who speaks the Egyptian dialect natively. In order to capture the L2-speaking characteristics and to build a dataset across dialects of Arabic, we have collected voice recordings from the same speaker in Egyptian (EGY), Modern Standard Arabic (MSA), English (ENG), and Egyptian-English code-switching (CS).

All of the data has been recorded in a noise-free environment at 22KHz sampling rate and in the 16-bit Signed Integer PCM audio encoding format. Each of the dialect(s)/language(s) involved has equal representation in the dataset. For each dialect, one hour of speech has been extracted after discarding segments with silences. For each dialect/language, the data has

then been split randomly into ~ 45 -min as training, ~ 10 -min as development, and ~ 5 -min as testing sets. A complete statistical breakdown of the dataset is available in Table 1.

While collecting voice-recordings from native speakers of the language is a common practice, to the best of our knowledge this is the first work which explores the collation of a TTS dataset which has recordings from the same speaker in the native language as well as other dialects and languages where he is an L2-speaker.

Most formal and informal conversations in urban settings worldwide involve code-switching between the native language and another language, which is often English. However, limited efforts have been made to collect data sets for downstream speech tasks to represent this trend. While providing Egyptian-English code-switching data, SawtArabi also offers the advantage of including mono-lingual Egyptian and English data, from the same speaker. We believe this approach to dataset collection can help training and understanding challenges with both mono-lingual and multi-lingual TTS systems.

3. Experimental Design

3.1. Vowelizer

The Arabic writing system includes 36 consonants and 8 diacritics [18]. However, most text is unvowelized, causing ambiguities that challenge TTS systems. Diacritization (aka vowelization) is essential for producing intelligible, natural-sounding speech, as missing vowels can lead to mispronunciations and degrade the naturalness of TTS [18]. For example, the word علم can be pronounced as /?alam/ (flag), /?ilm/ (knowledge), or /?allama/ (he taught), depending on the diacritics. To assess this effect, we performed two experiments: one with vowelized transcription of the dataset and the other without diacritics.

3.2. Phonemizer

Grapheme-to-Phoneme (G2P) conversion is crucial for transforming written text into phonetic representations. Although vowelized Arabic text captures most phonetic information, certain phonetic features are still better represented in phonetic transcription. For instance, assimilation affects pronunciation, as in the case of the word ذنب (*nb), where the ن (n) should be pronounced as م (m) (e.g., /?damb/ instead of /?dnb/).

We employ espeak-ng due to its widespread use and multilingual support, including Arabic. espeak-ng utilizes the International Phonetic Alphabet (IPA) and follows a rule-based approach combined with dictionary lookups, enabling easy customization and improved handling of out-of-vocabulary words³.

²Buckwalter transliteration for the Arabic text https://en.wikipedia.org/wiki/Buckwalter_transliteration

³<https://espeak.sourceforge.net/dictionary.html>

Despite its effectiveness in Arabic G2P conversion, *espeak-ng* still produces several mispronunciations. This section highlights key issues and the modifications made to improve its accuracy. Enhancements include refining the Arabic rule-based file and applying pre-processing and post-processing before phonemization. These modifications were implemented in *espeak-ng* version 1.52. To evaluate the effectiveness of the modified phonemizer, new experiments were conducted, comparing its results with the standard *espeak-ng* implementation.

3.2.1. Same Phonetic Symbol for two different Arabic letters

By default, *espeak-ng* assigns the same phonetic symbol /ð/ to both ط (Z) and ذ (*). To address this, the *ph_arabic* file was modified to assign /ðʔ/ to ط (Z). Although *espeak-ng* always places a stress mark after the vowel following ط (Z), which helps differentiate it from ذ (*), but it does not apply any distinction when no vowel follows.

3.2.2. Incorrect Pronunciation of Ta' Marbuta with Sukun

The letter *Ta' Marbuta* ð (p) should be pronounced as /h/ when carrying *Sukun* (no diacritics), but *espeak-ng* incorrectly retains it as /t/. This issue was resolved by modifying the *ar_rules* file and adjusting the rules of the pronunciation of ð (p).

3.2.3. Mispronunciation of Hamzat Al-Wasl in the definite article (Al- the)

espeak-ng assigns /ʔ/ to the *Hamzat Al-Wasl* (l) in the definite article (Al- the) (ال التعريف). This is only correct when the word appears at the beginning of a sentence (start of speech). However, in continuous speech, when the definite article (ال) occurs in the middle of a sentence, *Hamzat Al-Wasl* (l) should be silent. To address this, two modifications were applied:

1. A Python script was implemented to adjust the definite article (ال التعريف) based on its position in the sentence. Al-Qamariyyah (ال) is converted to أل (/ʔal/), while in Al-Shamsiyyah (ال), l is removed and l is replaced with ʔ (/ʔa/).
2. The glottal stop /ʔ/ was removed from the pronunciation rules of ال التعريف in the *ar_rules* file under the l group rules.

3.2.4. Standardizing Shaddah (Gemination) Representation

In IPA notation, Shaddah (gemination) can be represented in two ways: using a colon after the consonant or doubling the consonant (e.g., ss). As *espeak-ng* inconsistently alternates between these representations, a Python post-processing script has been implemented to standardize the geminate format by always using the duplicated letter representation, e.g. s: → ss. Additionally, in some cases, *espeak-ng* incorrectly produces three phonemes instead of two for geminated consonants, either as ss: or sss.

3.3. Modeling

In order to demonstrate the effectiveness of the dataset, we have trained baseline TTS systems on each of the dialect(s)/language(s). This involves training a Text-to-Spectrogram model and a vocoder for each of the dialect(s)/language(s). The following subsections elaborate on the model choices and the training setup.

3.3.1. TTS Training

Considering the non-autoregressive approach and the ability to train from scratch without the need for any external alignments, Matcha-TTS [19] has been chosen as the framework for training the baseline Text-to-MelSpectrogram models.

In order to study the effect of the standard *espeak-ng* phonemizer and the modified *espeak-ng* phonemizer 3.2, we train a different Matcha-TTS system on each of the phonemizers for the EGY, MSA and CS sections of our dataset.

While monolingual sentences have been phonemized by the standard *espeak-ng* or modified *espeak-ng* phonemizers directly by specifying the language, we follow a slightly different approach for the CS sentences. CS sentences have been split at phrase level with each phrase being completely monolingual. These monolingual phrases have then been phonemized by the respective phonemizers. The resulting phoneme sequence for a code-switching sentence comes from the phonemization of monolingual phrases that are embedded within.

Each TTS system has been trained on a single A-100 (80GB) GPU with a batch size of 256 for 16,399 epochs. Rest of the configurations follow the standard recipe from Matcha-TTS. Each of the resulting TTS systems have 18.2M parameters.

3.3.2. Vocoder Training

As HiFi-GAN [20] is among the most used vocoders and given that Matcha-TTS also supports the same, we have trained our vocoders on each of the dialect(s)/language(s). We have used the *config_v1* as a part of our work. This resulted in training 4 vocoder models using the standard training recipe from HiFi-GAN for 815000 updates.

4. Evaluation, Results and Discussion

4.1. Baseline evaluation

The baseline TTS systems have been evaluated on both objective and subjective metrics. Mel-cepstral distortion (MCD) and log- F_0 root mean square error (F_0 RMSE) have been used as the objective metrics. MCD as a measure captures how close two sequences of mel spectra are. The two sequences in this case are the ground truth and the synthesized speech. Log- F_0 RMSE on the other hand, refers to the logarithm of the root-mean-square error (RMSE) of the fundamental frequency (f_0) contour of the synthesized speech compared to the target f_0 contour of the reference speech.

Mean Opinion Score (MOS) has been used as the subjective measure to evaluate the perceived quality of speech samples by human listeners. This listening test has been designed to evaluate the intelligibility, naturalness, and pronunciation of the speech samples on a scale of 1 to 5. MOS score is then computed by averaging the scores assigned by multiple human listeners with a 95% confidence interval estimate. As a part of our evaluation, for each MOS test, we have presented 25 samples each from the Ground Truth and the speech synthesized by the corresponding TTS system. This subjective evaluation has been performed by 25 listeners proficient in different dialects, with English being their L2 language. These 25 sentences have been sourced from our test sets. As a result, each evaluator rates 25 samples of ground truth speech and 25 samples of the outputs from each of the TTS systems. Considering the different systems presented in Table 2, each evaluator has rated 275 audio samples in total.

Table 2 presents the evaluation results (objective and subjective) for the different TTS systems. While the objective metrics on the standard and modified phonemizers yield results that are quite similar, the subjective MOS scores suggest the significance

Table 2: Evaluation of the ground truth and baseline models over subjective and objective evaluation metrics. Intelligibility, Naturalness and Pronunciation are captured as MOS scores with 95% Confidence Interval (CI)

Model	Intelligibility	Naturalness	Pronunciation	MCD	F_0 RMSE
EGY (ground truth)	4.99 $\pm$ 0.01	4.96 $\pm$ 0.02	4.97 $\pm$ 0.02	-	-
EGY (standard espeak-ng)	4.75 $\pm$ 0.04	3.94 $\pm$ 0.08	4.43 $\pm$ 0.07	5.97 $\pm$ 0.70	0.23 $\pm$ 0.05
EGY (modified espeak-ng)	4.80 $\pm$ 0.04	4.11 $\pm$ 0.08	4.57 $\pm$ 0.06	5.95 $\pm$ 0.68	0.23 $\pm$ 0.05
MSA (ground truth)	4.9 $\pm$ 0.03	4.37 $\pm$ 0.08	4.87 $\pm$ 0.03	-	-
MSA (standard espeak-ng)	4.35 $\pm$ 0.08	3.03 $\pm$ 0.11	4.19 $\pm$ 0.08	4.86 $\pm$ 0.38	0.25 $\pm$ 0.06
MSA (modified espeak-ng)	4.39 $\pm$ 0.08	3.15 $\pm$ 0.11	4.27 $\pm$ 0.08	4.89 $\pm$ 0.37	0.24 $\pm$ 0.04
CS (ground truth)	4.99 $\pm$ 0.01	4.92 $\pm$ 0.03	4.97 $\pm$ 0.01	-	-
CS (standard espeak-ng)	4.55 $\pm$ 0.06	3.76 $\pm$ 0.09	4.19 $\pm$ 0.09	5.39 $\pm$ 0.69	0.24 $\pm$ 0.06
CS (modified espeak-ng)	4.60 $\pm$ 0.05	3.78 $\pm$ 0.09	4.23 $\pm$ 0.08	5.44 $\pm$ 0.62	0.25 $\pm$ 0.09
ENG (ground truth)	4.92 $\pm$ 0.02	4.71 $\pm$ 0.05	4.83 $\pm$ 0.04	-	-
ENG (standard espeak-ng)	4.77 $\pm$ 0.04	4.23 $\pm$ 0.07	4.56 $\pm$ 0.06	6.18 $\pm$ 0.74	0.28 $\pm$ 0.07

Table 3: Effect of vowelization (presence of diacritics) on the quality of TTS in MSA. Intelligibility and Pronunciation are captured as MOS scores with 95% Confidence Interval (CI)

Model	Intelligibility	Naturalness	Pronunciation
Ground Truth	4.97 $\pm$ 0.01	4.59 $\pm$ 0.06	4.93 $\pm$ 0.02
Without vowelization	3.2 $\pm$ 0.11	2.67 $\pm$ 0.1	2.53 $\pm$ 0.1
With vowelization	4.22 $\pm$ 0.09	3.21 $\pm$ 0.1	3.95 $\pm$ 0.09

of the modified phonemizer for Arabic.

4.2. Impact of the modified Arabic phonemizer

Table 2 presents the results comparing the standard espeak-ng with the modified version. The TTS systems built using the modified phonemizer consistently outperform the ones built using the standard phonemizer across all datasets and evaluation criteria, suggesting the need for better G2P conversion in Arabic. To further analyze the improvements, we examined the Arabic words flagged by evaluators as mispronounced. The standard espeak-ng evaluation has 157 mispronounced words (70 unique mispronunciations), while the modified version has 97 mistakes (44 unique cases). Both models shared 20 mispronounced words.

To better understand the impact of the modifications, we focused on the top 10 recurring mispronunciations in the standard model that were absent in the modified version. Among these, five errors were directly linked to issues in the standard phonemizer. The key findings are as follows:

- Two of the most frequent mispronunciations in the standard model involved the ignoring of ت (t) at the end of the word, as the TTS model skips it in the words شَارَفَت (\$aArafat) and كَلِمَات (kalimaAt). This issue arises because espeak-ng assigns /t/ to *Ta Marbuta* even when it lacks a vowel, whereas in natural speech, this /t/ is typically silent. The modified phonemizer, as discussed in Section 3.2.2, resolves this by replacing /t/ with /h/ where appropriate, ensuring that /t/ is only pronounced when required.
- A common issue, discussed in section 3.2.3, is the handling of the glottal stop at the beginning of words. The standard phonemizer consistently assigns /ʔal/ to the definite article (ال التعريف), even in contexts where it is typically silent in connected speech. As a result, instances where the glottal stop should be pronounced are often omitted. For example, in the word أَنْشِطَةُ (>an\$ITapu) appearing mid-sentence, the TTS system rendered it as /anʃiʔatu/ instead of the correct

/ʔanʃiʔatu/

- Another issue occurred with the pronunciation of geminated consonants, where the standard model produced an unnatural elongation of stress. This was due to espeak-ng generating three consecutive consonants instead of two, as addressed in section 3.2.4. Examples of this error include:
 - الرَّشَاوِي (Alrā\$Awiyy): Standard: /ʔarraʃa:wi/, Modified: /arraʃa:wi/
 - اللَّهُ (Allāh): Standard: /ʔallah/, Modified: /allah/

We analyzed the modified phonemizer to verify it introduced no new errors and found its phonemic representations remained consistent with the standard phonemizer.

4.3. Impact of vowelization

Table 3 presents the results of the experiment comparing vowelized and non-vowelized text in MSA. The two systems were trained using the standard espeak-ng phonemizer, and the same vocoder was used for both systems since only the TTS system training differs. As the table shows, the difference between the two experiments is significant across all criteria, particularly in pronunciation, highlighting the importance of vowelization for producing accurate and natural speech. By examining the phonemized text, we see that espeak-ng added vowels to certain words based on its Arabic dictionary, which helped improve the non-vowelized results. Listening to the generated audio revealed that missing vowels cause consonant clustering, making some phonemes hard to distinguish and reducing intelligibility. In MSA, omitted case-ending vowels further contribute to robotic and unnatural speech.

5. Conclusion and Future Work

In this paper, we presented a unique dataset consisting of single-speaker voice recordings in Egyptian Arabic, Modern Standard Arabic, English, and Egyptian-English code-switching. This is, to the best of our knowledge, the first study addressing dialectal and code-switching TTS in the Arabic context. Additionally, the dataset provides insights into the L2-speaking characteristics of a native Egyptian speaker. We also introduced modifications to the widely used espeak-ng phonemizer, enhancing the accuracy of Arabic phonemization, which improved the accuracy of TTS models. Our results also highlight the importance of vowelization. Furthermore, the dataset establishes a crucial benchmark for future TTS systems in Arabic.

As a part of the future work, we aim to expand the dataset by including more Arabic dialects and exploring the differences between dialectal and MSA writing conventions.

6. References

- [1] C. Wang, S. Chen, Y. Wu, Z. Zhang, L. Zhou, S. Liu, Z. Chen, Y. Liu, H. Wang, J. Li *et al.*, “Neural codec language models are zero-shot text to speech synthesizers,” *arXiv preprint arXiv:2301.02111*, 2023.
- [2] J. Kahn, M. Riviere, W. Zheng, E. Kharitonov, Q. Xu, P.-E. Mazaré, J. Karadayi, V. Liptchinsky, R. Collobert, C. Fuegen *et al.*, “Libri-light: A benchmark for ASR with limited or no supervision,” in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 7669–7673.
- [3] K. Shen, Z. Ju, X. Tan, Y. Liu, Y. Leng, L. He, T. Qin, S. Zhao, and J. Bian, “Naturalspeech 2: Latent diffusion models are natural and zero-shot speech and singing synthesizers,” *arXiv preprint arXiv:2304.09116*, 2023.
- [4] V. Pratap, Q. Xu, A. Sriram, G. Synnaeve, and R. Collobert, “Mls: A large-scale multilingual dataset for speech research,” in *Inter-speech 2020*, 2020.
- [5] L. Ma, D. Guo, K. Song, Y. Jiang, S. Wang, L. Xue, W. Xu, H. Zhao, B. Zhang, and L. Xie, “Wenetspeech4tts: A 12,800-hour mandarin tts corpus for large speech generation model benchmark,” in *Interspeech 2024*, 2024.
- [6] A. L. Thomas, A. Prakash, A. Baby, and H. Murthy, “Code-switching in indic speech synthesizers,” in *Interspeech 2018*, 2018, pp. 1948–1952.
- [7] S. Zhao, T. H. Nguyen, H. Wang, and B. Ma, “Towards natural bilingual and code-switched speech synthesis based on mix of monolingual recordings and cross-lingual voice conversion,” in *Interspeech 2020*, 2020, pp. 2927–2931.
- [8] X. Zhou, X. Tian, G. Lee, R. K. Das, and H. Li, “End-to-end code-switching tts with cross-lingual language model,” in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020.
- [9] K. Chen, Z. Huang, K. Lu, and Y. Yan, “Cosdiff: Code-switching tts model based on a multi-task ddim,” in *2024 IEEE International Conference on Multimedia and Expo (ICME)*, 2024.
- [10] M. Baali, T. Hayashi, H. Mubarak, S. Maiti, S. Watanabe, W. El-Hajj, and A. Ali, “Unsupervised data selection for tts: Using arabic broadcast news as a case study,” *arXiv preprint arXiv:2301.09099*, 2023.
- [11] A. Kulkarni, A. Kulkarni, S. A. M. Shatnawi, and H. Aldarmaki, “Clartts: An open-source classical arabic text-to-speech corpus,” in *Interspeech 2023*, 2023.
- [12] H. O. Toyin, A. Djanibekov, A. Kulkarni, and H. Aldarmaki, “ArTST: Arabic text and speech transformer,” in *Proceedings of ArabicNLP 2023*, H. Sawaf, S. El-Beltagy, W. Zaghouani, W. Magdy, A. Abdelali, N. Tomeh, I. Abu Farha, N. Habash, S. Khalifa, A. Keleg, H. Haddad, I. Zitouni, K. Mrini, and R. Al-matham, Eds. Association for Computational Linguistics, Dec. 2023.
- [13] A. Abdelali, N. Durrani, C. Demiroglu, F. Dalvi, H. Mubarak, and K. Darwish, “NatiQ: An end-to-end text-to-speech system for Arabic,” in *Proceedings of the Seventh Arabic Natural Language Processing Workshop (WANLP)*, H. Bouamor, H. Al-Khalifa, K. Darwish, O. Rambow, F. Bougares, A. Abdelali, N. Tomeh, S. Khalifa, and W. Zaghouani, Eds. Association for Computational Linguistics, 2022. [Online]. Available: <https://aclanthology.org/2022.wanlp-1.38/>
- [14] R. Yamamoto, E. Song, and J.-M. Kim, “Parallel wavegan: A fast waveform generation model based on generative adversarial networks with multi-resolution spectrogram,” in *ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020.
- [15] K. Ito and L. Johnson, “The lj speech dataset,” <https://keithito.com/LJ-Speech-Dataset/>, 2017.
- [16] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat *et al.*, “Gpt-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [17] S. A. Chowdhury, Y. Samih, M. Eldesouki, and A. Ali, “Effects of dialectal code-switching on speech modules: A study using egyptian arabic broadcast speech.” 2020.
- [18] L. Gu, W. Zhang, L. Tahir, and Y. Gao, “Statistical vowelization of arabic text for speech synthesis in speech-to-speech translation systems,” in *Interspeech*, 2007. [Online]. Available: <https://api.semanticscholar.org/CorpusID:31469389>
- [19] S. Mehta, R. Tu, J. Beskow, É. Székely, and G. E. Henter, “Matcha-TTS: A fast TTS architecture with conditional flow matching,” in *Proc. ICASSP*, 2024.
- [20] J. Kong, J. Kim, and J. Bae, “Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis,” *Advances in neural information processing systems*, vol. 33, pp. 17 022–17 033, 2020.